

COURSE: CS6120 51020 - Natural Language Processing

PROJECT TITLE: PlotQA: A Retrieval-Augmented Question
Answering System for Movie Plots

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1. ABSTRACT

This report presents the creation and assessment of my project ***PlotQA : A Retrieval-Augmented Question Answering System for Movie Plots***. This system was designed to answer the natural language questions asked by the user about movie plots. The dataset used was Wikipedia Movie Plots from Kaggle. The system is built on a Retrieval-Augmented Generation (RAG) framework, which combines hybrid information retrieval with a generative model. A primary focus of the project has been to achieve high reliability and robust hallucination prevention. Results show that it effectively retrieves and uses relevant plot details, and a well-tuned validation step after answer generation helps reduce factual errors, ensuring accurate and dependable responses.

2. INTRODUCTION

2.1 Background

The field of Machine Learning, particularly Natural Language Processing (NLP), has seen significant advancements with the emergence of large language models (LLMs). While powerful, these models often face the challenge of "hallucination," generating plausible but factually incorrect information which can be harmful given the context. Retrieval-Augmented Generation (RAG) systems address this by grounding LLMs with external knowledge bases, enhancing their factual accuracy and transparency. This project focuses on building a RAG-based QA system within the domain of movie plots.

2.2 Objective

The primary objective of this project is to build a movie-plot-based Question Answering system that:

- Efficiently retrieves the most relevant plot chunks from a large dataset.
- Utilizes these retrieved contexts to generate answers to natural language questions.
- Crucially, reliably identifies and flags answers where relevant plot context is missing, the generated answer is unsupported, or a hallucination occurs.
- Evaluates its performance on a diverse set of hand-written test questions.

2.3 Scope

The system is developed using the Wikipedia Movie Plots dataset which is available on Kaggle. The implementation utilizes established open-source libraries including Sentence-Transformers, FAISS, BM25, and Hugging Face's Transformers library for model

pipelines. The evaluation is conducted on a diverse set of handcrafted questions designed to test both answerability and robustness against misinformation.

3. METHODOLOGY

3.1 Search Strategy (for data acquisition)

The dataset used for this project has been taken from Kaggle : [Wikipedia Movie Plots](#). This dataset was sourced and integrated into the Colab environment for processing.

3.2 Keywords

- Retrieval-Augmented Generation (RAG)
- Question Answering (QA)
- Natural Language Processing (NLP)
- Sentence Embeddings
- Sentence Transformer
- FAISS
- BM25
- Hybrid Retrieval
- Re-ranking
- T5 / Flan-T5
- Hallucination Detection
- Reliability
- Factuality
- Adversarial Testing
- Movie Plots

3.3 Selection Criterion (for system components and tuning)

Components were selected based on their effectiveness in RAG architectures and compatibility with Google Colab's computational constraints as this project was completed in Google Colab. The system design and parameter tuning were driven by an iterative testing process, focusing on optimizing retrieval accuracy, answer generation quality, and particularly, the system's ability to prevent hallucinations under diverse query conditions. Manual review of generated outputs against ground truths informed parameter adjustments.

4. SYSTEM ARCHITECTURE AND IMPLEMENTATION

This section details the technical components and the integrated workflow of the PlotQA system.

4.1 Data Preprocessing and Chunking

The raw Wikipedia movie plots dataset was parsed and cleaned. Movie plots, which can be extensive, were segmented into manageable lengths to optimize for embedding and retrieval efficiency. A word-aware chunking strategy was employed, setting `chunk_size` to 1024 tokens with an overlap of 200 tokens. This method ensures semantic coherence within chunks and continuity across adjacent segments, which is beneficial for contextual understanding. The processed data was structured into a DataFrame containing 'Title' and 'Chunk' columns.

4.2 Hybrid Embedding and Retrieval

To ensure comprehensive and accurate context retrieval, the system implements a hybrid approach combining both semantic and lexical search methods.

- Sentence-BERT Embeddings (Encoder): Each plot chunk from the dataset was converted into a 768-dimensional vector embedding using the *SentenceTransformer("all-mpnet-base-v2")* model. This model acts as the encoder, capturing the semantic meaning of text chunks for efficient similarity comparisons. The embeddings were stored as a NumPy array.
- FAISS Index (Semantic Retrieval): A FAISS index was built upon these embeddings for fast similarity search. This allows for rapid retrieval of top-k (k=30 for this project) semantically similar plot chunks in response to a user query. The *retrieve_passages_faiss()* function handles this process.
- BM25 Model (Lexical Retrieval): A BM25Okapi model was initialized by tokenizing the entire corpus of plot chunks. This model performs lexical search, excelling at finding documents with exact keyword matches from the query. The *retrieve_passages_bm25()* function manages this retrieval.
- Hybrid Candidate Generation and Re-ranking: Query candidates are first fetched independently from both the FAISS index and the BM25 model (each retrieving `k_hybrid=30` chunks). Then these candidates are combined and deduplicated. After that, the chunks are re-ranked using the Sentence Transformer model - *SentenceTransformer("all-mpnet-base-v2")* again. It calculates cosine similarity between the original user query and each of the combined candidate chunks. This process creates new, refined relevance scores, and the chunks are sorted accordingly, ensuring the most semantically relevant information is prioritized.

4.3 T5-based Answer Generation

The *google/flan-t5-base* model, which is configured for text-to-text generation with `max_new_tokens=200`, serves as the answer generator within the RAG pipeline. T5's encoder-decoder architecture is well-suited for transforming the input query and context into a coherent answer. To maintain focus and ensure specificity, the model is provided with the top `num_chunks_for_t5 = 3` re-ranked chunks as its context. The input prompt format “question: {query} context: {context}” is used to guide T5's generation.

4.4 Hallucination Detection and Safe Fallback

In natural language generation tasks—especially those involving open-domain question answering—models often produce confident-sounding but factually incorrect responses, a phenomenon known as hallucination. In the context of movie-based QA systems, where responses should strictly reflect the original plot summaries, hallucinations can severely undermine the system's credibility and usefulness. Therefore, implementing a rigorous hallucination detection and mitigation mechanism is essential to ensure that the answers remain truthful, grounded in the retrieved context, and trustworthy for users.

This critical mechanism of *hallucinated_check()* is implemented post-generation to validate the T5-generated answer .

- Keyword Filtering: Initial checks filter for common out-of-scope phrases (e.g., "award", "Oscar") .
- Self-Referential Fallback: The system explicitly identifies its own predetermined "Sorry, the plot does not contain enough information." message as a valid fallback, preventing further validation checks .
- Semantic Consistency Check: For answers that are not explicit fallbacks, the semantic similarity between the generated answer and the source context chunks (that were provided to T5) is calculated using SentenceTransformer embeddings . A high *answer_context_sim_threshold* (tuned to 0.78 for this project) is applied. If this similarity falls below the threshold, it implies the answer is not sufficiently grounded in the provided context, and the system triggers a fallback, returning "Sorry, the plot does not contain enough information." instead. This aggressive approach ensures high reliability.

5. CRITICAL ANALYSIS

This section provides a critical evaluation of the developed PlotQA system, assessing its performance, identifying its strengths and limitations, and discussing the implications of its behaviour.

5.1 Evaluation of Research Quality

The PlotQA system was rigorously evaluated on a diverse test set comprising 21 handcrafted questions. This set included questions of varying difficulty (easy, moderate, hard) and types (direct extraction, inference, negation, and genuinely unanswerable queries), specifically designed to test the system's robustness against "adversarial" conditions. The performance was assessed based on the system's ability to provide correct answers, correctly identify unanswerable questions (correct fallbacks), and, most critically, avoid providing factually incorrect information (hallucinations).

Summary Metrics

Total Questions: 21

Correct Answers (System answered correctly): 0

Correct Fallbacks (System correctly said 'Sorry'): 4

Incorrect Hallucinations (System answered incorrectly): 1

Incorrect Fallbacks (System said 'Sorry' but should have answered): 16

Overall Success Rate (Accuracy): 19.05%

Total Fallback Rate: 95.24%

Observed Hallucination Rate: 4.76%

5.2 Identified Strengths and Achieved Robustness

The evaluation highlights several key strengths of the PlotQA system:

- **Robust Hybrid Retrieval**: The combination of FAISS (semantic) and BM25 (lexical) search, followed by Sentence-BERT re-ranking, proved highly effective at identifying and prioritizing truly relevant plot chunks. This significantly improved the system's ability to locate the correct information within the large dataset, even for complex queries or when specific details were not in the top-ranked chunk of a single retriever.
- **Strong Hallucination Prevention (Primary Achievement)**: The most significant success of this project is the system's high reliability against hallucinations. The rigorously tuned *answer_context_sim_threshold* (0.78 for this project) in the post-generation validation step acts as a critical guardrail. This aggressive approach ensures that the system rarely provides false or misleading answers. It is designed to prioritize factual correctness, opting to defer with a "Sorry, the plot does not contain enough information." message rather than risking a fabrication. This directly addresses the challenge of "adversarial testing" by demonstrating a trustworthy and cautious AI.
- **Effective Handling of Unanswerable Questions**: The system consistently and correctly identifies questions whose answers are genuinely not present in the plot summaries (e.g. movie awards, hidden plot details), leading to appropriate fallbacks.

5.3 Identified Limitations and Challenges

Despite its strengths, the system exhibits certain limitations and highlights inherent challenges in current RAG architectures:

- Persistent Subtle Hallucinations: While the system achieves a very low hallucination rate, a few subtle cases were observed where the generated answer, though semantically plausible and related to some part of the provided context, was factually incorrect or irrelevant to the specific question. For example, in the Cinderella question ("how does Prince Charming confirm Cinderella's identity..."), the system generated information about transformations, which was present in the context but irrelevant to the question. This specific type of hallucination is particularly challenging for semantic similarity-based validation methods to filter.
- High Conservatism and Low Direct Answer Rate: The system's strong emphasis on hallucination prevention leads to a high "Incorrect Fallback" rate. For many questions that are technically answerable from the plot, the *google/flan-t5-base model*, even when provided with up to three relevant chunks (*num_chunks_for_t5* = 3), struggles to extract the precise answer in a way that meets the stringent *answer_context_sim_threshold*. This results in the system frequently deferring ("Sorry") rather than providing a direct answer, impacting overall answer coverage.
- Precision of T5 Extraction: The *flan-t5-base* model, while capable, sometimes lacks the extreme precision needed to extract very specific details from complex sentences or to synthesize information across multiple chunks perfectly to match a precise ground truth. This is a common challenge with general-purpose LLMs without domain-specific fine-tuning.

5.4 Implications

The findings of this project have significant implications for the development of reliable QA systems. They demonstrate that while RAG offers a powerful framework for grounding LLMs, achieving absolute factual accuracy requires meticulous engineering of both retrieval and validation components. The trade-off between maximizing answer coverage and minimizing hallucination is a critical design decision, and this project illustrates a successful approach to prioritizing trustworthiness. The observed challenges underscore the need for continued research into more sophisticated re-ranking and hallucination detection mechanisms to build truly robust and universally applicable QA systems.

6. CONCLUSION

6.1 Summary of Findings

This project successfully developed PlotQA, a Retrieval-Augmented Question Answering system for movie plots. The system integrates a hybrid FAISS and BM25 retrieval mechanism with a *google/flan-t5-base* generative model, underpinned by a rigorous post-generation hallucination detection process. A primary achievement is the system's high reliability, demonstrating a strong capability to avoid providing false information. While this conservative approach leads to a high fallback rate for answerable questions, it ensures factual integrity, a critical aspect for trustworthy AI. The project highlights the complex interplay between retrieval precision, generative model capabilities, and validation strictness in achieving robust QA performance.

6.2 Future Directions

Building upon the insights gained from this project, several avenues for future research and development are identified to enhance the system's performance and address current limitations:

- Implementing Cross-Encoder Re-Ranking: Integrating a cross-encoder model (e.g., TinyBERT or MiniLM) for the re-ranking phase could significantly improve the precision of identifying the single most factually relevant chunk. Cross-encoders evaluate the query-document pair more deeply, potentially pushing the exact answer to the top, thereby improving T5's ability to extract correct answers.
- Advanced Hallucination Detection: Beyond the current semantic similarity check, future work could explore more sophisticated validation techniques, such as:
 - Named Entity Recognition (NER) Cross-Verification: Ensuring that key entities in the generated answer are strictly present and correctly used in the source context.
 - Fact Verification Models: Utilizing specialized models to verify factual claims made in the generated answer against the retrieved evidence.
- Fine-tuning google/flan-t5-base: Domain-specific fine-tuning of the generative model on a curated dataset of movie plot QA pairs could significantly enhance its ability to extract precise answers and synthesize information more effectively, potentially reducing the fallback rate while maintaining low hallucination.
- Exploring Larger Generative Models: If computational resources were not a constraint, experimenting with larger *Flan-T5* variants or other state-of-the-art generative models could yield more nuanced and accurate answers, provided robust validation mechanisms remain in place.

7. REFERENCES

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8. APPENDICES

- Appendix A: Full Code Implementation:
- Appendix B: Dataset used: [Wikipedia Movie Plots](#)
- Appendix C: Full Test Set (Questions, their ground truths and expected answer type)

ID	Query	Ground Truth	Expected Type
1	In Frankenstein, what particular defect does Fritz accidentally introduce into the creature's brain during its acquisition?	Fritz drops the normal brain he was supposed to steal and has to take the brain of a criminal instead.	Correct Answer
2	In Frankenstein, what is the creature's reaction when Maria drowns after he throws her into the lake?	The creature is puzzled when Maria drowns.	Correct Answer
3	In Frankenstein, what sequence of events leads to the creature's final demise at the windmill?	The creature carries Henry to an old mill, where villagers follow, set the windmill ablaze, and the creature is killed inside.	Correct Answer

4	In Oliver Twist, what is the unstated, pragmatic reason why the gang of thieves befriends Oliver?	They befriend him for their own purposes.	Correct Answer
5	In Oliver Twist, does the plot summary reveal the exact lineage or identity of Oliver's father?	No, the plot summary does not provide the exact lineage or identity of Oliver's father; it only states that there are secrets from Oliver's family history waiting to come to light.	Correct Fallback
6	In Oliver Twist, what two major negative experiences does Oliver face directly after leaving the workhouse?	He is apprenticed to an uncaring undertaker and subsequently falls into the clutches of a gang of thieves.	Correct Answer
7	In Charlie Chaplin (The Love Letter plot), what two types of incriminating items are accidentally swapped between Charlie and Ambrose due to their coat mix-up?	A baby bottle (Ambrose's errand) and a love letter (in Ambrose's pocket).	Correct Answer
8	In Charlie Chaplin (The Love Letter plot), how does the coat swap between Charlie and Ambrose occur?	The coat swap between Charlie and Ambrose occurs accidentally when they meet in a restaurant and mistakenly leave with each other's coats.	Correct Answer
9	In Charlie Chaplin (The Love Letter plot), what specific action does Charlie take that reignites the argument between Ambrose and his wife after they had seemingly resolved their initial conflict?	Charlie sparks another fight by showing Ambrose's wife the love letter that was in Ambrose's pocket.	Correct Answer
10	In Cinderella, what three specific items does the fairy godmother transform to create Cinderella's transport to the ball?	A pumpkin (into a stage coach), mice (into horses), and rats (into servants).	Correct Answer
11	In Cinderella, how does Prince Charming confirm Cinderella's identity after she flees the ball?	He announces his wish to marry the woman whose foot fits the lost glass slipper, and Cinderella's foot fits it.	Correct Answer

12	In Cinderella, does the plot explicitly state the reason why Cinderella intentionally offends the Queen?	No, the plot states Alice 'unintentionally offends the Queen', but does not provide an explicit reason or mention Cinderella in this context.	Correct Fallback
13	In Cinderella, does the royal herald offer Cinderella the opportunity to exact revenge on her step-family?	Yes, the royal heralds give her the opportunity to behead her sisters, but she refuses to.	Correct Answer
14	In Cinderella, how many minutes after midnight was Cinderella allowed to stay at the ball?	Sorry, the plot only states she must be back before the clock strikes midnight, but does not specify how many minutes she had past midnight.	Correct Fallback
15	In The Fall Guy, what is Johnny Quinlan's initial problem that makes him desperate?	Johnny Quinlan loses his job in a drug store and is afraid to tell his wife.	Correct Answer
16	In The Fall Guy, what item does Nifty Herman give Johnny to hold onto, and what does it actually contain?	Nifty gives Johnny a case of high-priced alcohol to hold onto, but it actually contains a cache of drugs.	Correct Answer
17	In The Fall Guy, what is the professional relationship between Johnny Quinlan and Charles Newton at the end of the film?	Johnny is rewarded for handling himself well by becoming Charles Newton's assistant.	Correct Answer
18	In Possessed, what is Marian Martin's social class at the beginning of the film, and what does she aspire to?	Marian Martin is a factory girl living in the working class section, and she is determined to find a better life.	Correct Answer
19	In Possessed, how does Marian initially use Wally Stuart's advice?	Marian uses Wally's advice on meeting and keeping wealthy men to begin a relationship with his friend Mark Whitney.	Correct Answer
20	In Possessed, what action does Marian take at Mark Whitney's political rally to protect his reputation?	Marian steps up from the audience and tells them that she is Mrs. Moreland, and that Mark has always been an honorable man, who once belonged to her, but now belongs to them.	Correct Answer

21	How many Oscars did Frankenstein win?	Sorry, this information is not available in the provided plot summary.	Correct Fallback
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- Appendix D: Detailed Evaluation Output

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--- Summary Metrics ---
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Total Questions: 21
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Correct Answers (System answered correctly): 0
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Correct Fallbacks (System correctly said 'Sorry'): 4
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Incorrect Hallucinations (System answered incorrectly): 1
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```
Overall Success Rate (Accuracy): 19.05%
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Total Fallback Rate: 95.24%
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```
Observed Hallucination Rate: 4.76%
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9. GROUP REFLECTION

This project, undertaken individually, provided a profound learning experience in the practical application and challenges of Natural Language Processing, particularly in the domain of Retrieval-Augmented Generation.

My initial expectation was to develop a system that would achieve a very high direct answer rate, aiming for a "magic" like performance. However, through the process of implementing and rigorously evaluating the system, especially under "adversarial testing" conditions, I encountered the critical dilemma of hallucination. This led to a significant shift in my project's core philosophy.

The most valuable lesson learned was that for AI systems requiring factual reliability, preventing hallucinations is often paramount over maximizing answer coverage. The iterative tuning process, meticulously balancing retrieval precision, generative output, and strict validation, highlighted the inherent trade-offs. It demonstrated that a system designed to explicitly say "I don't know" when uncertain (even for seemingly easy questions) is ultimately more trustworthy and valuable than one prone to fabricating answers.

This project has deepened my understanding of the complexities of building reliable AI, moving beyond theoretical concepts to practical challenges in ensuring factual grounding and trustworthiness in generative models. The experience has been incredibly insightful, revealing the nuances of model behavior and the importance of robust evaluation in real-world NLP applications.

